

AEGIS: Matrix-Free Diagnostics for the Adversarial Fault Lines of Graph Neural Networks

Anonymous Author(s)

Abstract—Graph neural networks now inform safety-critical decisions in drug-interaction screening, fraud detection, and power-grid contingency analysis, where adversarial perturbations of the graph can flip predictions. Auditing such a model means mapping its *adversarial fault lines*: the most sensitive first-order perturbation direction, per-edge equilibrium-sensitivity rankings, and per-node sensitivity radii. No prior tool supplies all three from one object. AEGIS does so, via the *constrained sensitivity matrix* S_c , computed matrix-free through a Neumann-series resolvent and randomized SVD that scale to $N=7,650$ on one GPU. For the spectral-norm-constrained IGNN subclass we prove a closed-form three-regime characterisation with critical budget $\varepsilon_{\text{crit}}=(1-\kappa)/\|W\|_2$ (rate sharp in the all-active case), which trained models satisfy with a $2\text{--}4\times$ spectral-radius margin. The construction extends to any continuous-edge-weight message passing (GCN, SAGE, GIN, APPNP, IGNN, edge-weighted GAT²): its continuous-to-discrete bridge is positive in *all* 39/39 cells under the edge-weighted ranking of Proposition 3 (median $\tau=+0.99$, $p<10^{-5}$), beating an edge-weight baseline by $\Delta\tau$ up to $+0.65$ on a fraud graph and reaching $+0.996$ on Amazon Photo ($N=7,650$), while one query delivers $74\text{--}156\times$ the per-query damage of 50-step PGD. We further find adversarial vulnerability is *delocalized*: the leading attack mode spans tens of edges, so v_{ij} ranks risk far above chance yet few-edge masking gives only limited protection. A fraud-detector case study shows the same one-query audit in action (section VI).

Index Terms—Graph neural networks, adversarial robustness, constrained sensitivity, implicit graph neural networks, randomized SVD, fraud detection

I. INTRODUCTION

Graph neural networks are now deployed in domains where structural errors carry tangible consequences: fraudulent accounts evade detection via perturbed transaction edges [1], drug-interaction models misfire under perturbed molecular graphs [2], and power grids can miss critical contingencies when their graph representations are structurally fragile [3]. A practitioner deploying a GNN faces a question current tools leave unanswered: *which edges, if perturbed, would cause predictions to fail, and by how much?*

Each existing thread maps only part of these *adversarial fault lines*. Structural attacks [1], [4], [5], [6] rank edges by surrogate gradients but yield no continuous direction and no per-node budget; smoothing [7], [8], [9], [10] and IBP-style certifiers [11], [12] return per-node certificates but no edge ranking or direction; robust-architecture defenses [13], [14], [15] harden models without surfacing per-edge vulnerability; and sober-look critiques [16], [17] flag evaluation pitfalls. No prior method returns all three from a single matrix-free pass. AEGIS (Adversarial Evaluation of Graph Integrity via Sensitivity) does, from one analytical object: the constrained

sensitivity matrix S_c (section IV). Intuitively, S_c records how strongly each edge can push the model’s equilibrium prediction, so its leading singular direction, its column norms, and its per-node margins yield the three diagnostics in a single pass. Here “one query” means one matrix-free S_c construction (a single randomized SVD over the equilibrium resolvent), not three separate analyses. The closed-form regime characterisation (Theorem 1) is established for contractive implicit GNNs, and S_c extends to any GNN with continuous edge-weight-modulated message passing (section V-F); the threat model targets edge deletion and weight perturbation (section II).

Contributions. (1) **Constrained sensitivity operator.** S_c specialises equilibrium IFT sensitivity [18], [19] to *structural* edge perturbations via the projection P_c . One matrix-free query then reads the SVD-optimal direction, per-edge rankings, and per-node radii together, where no prior object yields all three; it matches a dense σ_1 to 0.03% at $N=200$ and scales to $N=7,650$. (2) **Phase-transition theory.** A worst-case three-regime characterisation with closed-form critical budget $\varepsilon_{\text{crit}}=(1-\kappa)/\|W\|_2$ for the spectral-norm-constrained IGNN subclass. Trained IGNNs hold a $2\text{--}4\times$ spectral-radius margin to criticality across our κ_{max} sweep, a closed-form structural-perturbation safety characterisation at $\sim 6\%$ accuracy cost (Theorem 1 and section V-D). (3) **Empirical evaluation.** 6 datasets, 7 architectures, 4 domains (390 runs): four-quadrant attack comparison, head-to-head vs. GR-BCD/PR-BCD [5] and AGNNCert [11] (table III), continuous-to-discrete transfer (fig. 7), and a fraud-detector audit (section VI).

II. PRELIMINARIES AND THREAT MODEL

Let $G = (V, E, A)$ be an undirected graph with $N = |V|$ nodes and adjacency $A \in \mathbb{R}^{N \times N}$, node features $X \in \mathbb{R}^{N \times d_{\text{in}}}$, and symmetric normalized adjacency $\hat{A} = D^{-1/2}(A+I)D^{-1/2}$ (self-loops included). Throughout, ϕ is the activation, $\sigma_i(\cdot)$ the i -th singular value, and $\text{vec}(\cdot)$ the column-major vectorization. The sensitivity matrices (S , the constrained S_c , the unrolled S_K , and the per-node block S_v) and the scalars that govern them (contractivity $\kappa = \|J_z\|_2$, spectral radius $\rho(J_z)$, pseudospectral index $\eta \geq 1$, critical budget $\varepsilon_{\text{crit}}$, per-node radius r_v) are each defined where they first appear.

IGNN and IFT. AEGIS analyses models that settle to an equilibrium, because that fixed point is what exposes structural sensitivity in closed form. A DEQ-GNN [20], [21] iterates a weight-tied F_θ to a fixed point; the IGNN instantiation [22] uses

$$F(Z, A) = \phi(\hat{A}ZW^\top + X_{\text{proj}}), \quad (1)$$

with $W \in \mathbb{R}^{d \times d}$ spectral-normalized [23] so $\kappa \leq \|\hat{A}\|_2 \|W\|_2 < 1$, ReLU ϕ , and a learned projection X_{proj} . Training uses implicit differentiation through the fixed point [19], [24] with optional Jacobian regularization [25], and a linear head $f(z^*) = W_{\text{cls}} z^* + b$ produces class predictions. At equilibrium, $G(z^*, A) = z^* - F(z^*, A) = 0$ defines z^* as an implicit function of A , with IFT sensitivity

$$\frac{\partial z^*}{\partial p} = (I - J_z)^{-1} \frac{\partial F}{\partial p} \Big|_{z^*}. \quad (2)$$

Equation (2) is the lever AEGIS pulls: it gives, in closed form, the first-order change in the equilibrium z^* under any parameter p , with the resolvent $(I - J_z)^{-1}$ amplifying a perturbation by at most $1/(1 - \kappa)$ through its convergent Neumann series. The pseudospectral index $\eta = \|(I - J_z)^{-1}\|_2 (1 - \rho) \geq 1$ [26] measures the gap to the spectral-radius bound, equalling 1 for normal J_z . The IFT has been used for hyperparameter optimization [27] and implicit differentiation [19], but not for adversarial structural analysis.

Threat model. A white-box attacker with full access to the trained IGNN perturbs the normalized adjacency

$$\hat{A}' = \hat{A} + \delta \hat{A}, \quad \|\delta \hat{A}\|_F \leq \varepsilon, \quad (3)$$

with $\delta \hat{A}$ symmetric ($\delta \hat{A} = \delta \hat{A}^\top$, preserving the undirected graph), edge-only ($[\delta \hat{A}]_{ij} = 0$ for $(i, j) \notin E$, so no new edges), and continuous in $\mathbb{R}$. AEGIS targets edge-deletion / weight-perturbation; insertion attacks (Nettack-class [1], [4] and node-injection [28]) require enlarging the basis to $\bar{E} \supseteq E$, a natural follow-up that preserves the S_c construction. Discrete deletions connect to the continuous model through Proposition 3. Unlike input-feature perturbation [29], [30], structural perturbation changes the operator itself: z^* depends on A through both $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ and $J_A = \partial F / \partial \text{vec}(A)$, motivating S_c in section III.

III. ADVERSARIAL EQUILIBRIUM THEORY

Building on the threat model of section II, we derive the analytical machinery behind the three diagnostics promised in section I: per-edge rankings, the SVD-optimal direction, and per-node radii. The constrained sensitivity matrix S_c is the unified object from which all three follow.

The governing picture is a phase transition in the perturbation budget ε . While ε stays small, the perturbed model remains contractive and its prediction moves smoothly and boundedly with the graph; as ε approaches a critical budget $\varepsilon_{\text{crit}}$, the equilibrium grows arbitrarily sensitive; past $\varepsilon_{\text{crit}}$, contraction is no longer guaranteed and the prediction can break. Theorem 1 makes this precise and puts $\varepsilon_{\text{crit}}$ in closed form.

Theorem 1 (Vulnerability Characterization for Contractive Implicit GNNs). *Let $F_\theta(Z, \hat{A}) = \phi(\hat{A}ZW^\top + X_{\text{proj}})$ be an IGNN-class operator satisfying:*

- (A1) ϕ is ReLU or any 1-Lipschitz activation; nonsmoothness at zero is handled via the conservative IFT [31] (proof below).
- (A2) $\|W\|_2 \leq c$ via spectral normalization.

- (A3) $\|J_z\|_2 \leq \kappa < 1$ verified post-training ($\kappa=0.14\text{--}0.59$ across all benchmarks in our suite; section V-E). (A2) does not imply (A3) automatically; we report the trained $\kappa = \|J_z\|_2$ alongside $\varepsilon_{\text{crit}}$ (worst case $\|\hat{A}\|_2 \|W\|_2$) so practitioners can audit, and AEGIS continues to return S_c rankings and SVD-optimal attacks if (A3) is not satisfied, with the formal guarantees below then scoped to the contractive regime.

Define the critical budget

$$\varepsilon_{\text{crit}} = \frac{1 - \kappa}{\|W\|_2}. \quad (4)$$

Then structural perturbations $\delta \hat{A}$ with $\|\delta \hat{A}\|_F = \varepsilon$ exhibit three regimes:

(a) **Subcritical** ($\varepsilon < \varepsilon_{\text{crit}}$): The perturbed operator remains contractive with a unique fixed point satisfying

$$\|\Delta z^*\|_F \leq \sigma_1(S) \cdot \varepsilon + O(\varepsilon^2), \quad (5)$$

where $S = (I - J_z)^{-1} J_A$ is the structural sensitivity matrix and $\sigma_1(S) \leq \|J_A\|_{\text{op}} / (1 - \kappa)$.

(b) **Critical** ($\varepsilon \rightarrow \varepsilon_{\text{crit}}$): The resolvent obeys the unconditional eigenvalue bound $\|(I - J'_z)^{-1}\|_2 \geq 1 / \min_i |1 - \lambda_i(J'_z)|$, so it blows up precisely when an eigenvalue of J'_z approaches 1: a vanishing spectral gap to 1, not $\|J'_z\|_2 \rightarrow 1$ on its own. The rate is $\Omega(1/(\varepsilon_{\text{crit}} - \varepsilon))$ when J'_z is normal with a dominant real-positive eigenvalue (the Perron mode of symmetric $\hat{A}, W$); in general $\varepsilon_{\text{crit}}$ is the norm-based contraction radius that lower-bounds the eigenvalue divergence threshold, the slack reflecting the nonnormality of J'_z (bounded by η , Observation 1; $\eta \leq 2.47$ for general ReLU, Remark 1).

(c) **Supercritical** ($\varepsilon \geq \varepsilon_{\text{crit}}$): The sufficient contraction certificate no longer applies ($\|J'_z\|_2$ may exceed 1) and the part-(a) first-order guarantees do not apply; at the worst-case $\kappa = \|\hat{A}\|_2 \|W\|_2$, $\varepsilon < \varepsilon_{\text{crit}}$ is a sufficient safety boundary that trained IGNNs sit well inside (a $2\text{--}4\times$ spectral-radius margin to $\rho=1$ across the κ_{max} sweep of section V-D).

Proof. (a). Under (A1), F_θ is piecewise-affine on each ReLU linear region; the activation pattern at z^* is generically stable for sufficiently small $\|\delta \hat{A}\|$ (the boundary set $\{\hat{A} : z^*(\hat{A}) \text{ lies on a region face}\}$ has Lebesgue measure zero), so the conservative-gradient calculus of Bolte–Pauwels [31] yields a conservative IFT on each region. With $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ and $\kappa \leq \|\hat{A}\|_2 \|W\|_2$, applying IFT to $G(z, \hat{A}) = z - F(z, \hat{A}) = 0$ at $(z^*, \hat{A})$ gives

$$\Delta z^* = (I - J_z)^{-1} J_A \cdot \text{vec}(\delta \hat{A}) + O(\|\delta \hat{A}\|^2), \quad (6)$$

with $J_A = \partial F / \partial \text{vec}(A)$ and Neumann $\|(I - J_z)^{-1}\|_2 \leq 1/(1 - \kappa)$ yielding (5). For contractivity preservation, $\|J'_z\|_2 \leq (\|\hat{A}\|_2 + \varepsilon) \|W\|_2$, so $\varepsilon < \varepsilon_{\text{crit}}$ guarantees $\|J'_z\|_2 < 1$.

(b)–(c). For any M , $(I - M)^{-1}$ has eigenvalues $(1 - \lambda_i(M))^{-1}$, hence $\|(I - M)^{-1}\|_2 \geq \rho((I - M)^{-1}) = 1 / \min_i |1 - \lambda_i(M)|$ unconditionally. Along the worst-case direction $\|J'_z\|_2 \leq (\|\hat{A}\|_2 + \varepsilon) \|W\|_2$, so

$1 - \|J'_z\|_2 \geq \|W\|_2(\varepsilon_{\text{crit}} - \varepsilon)$ and, since $\rho \leq \|\cdot\|_2$, $\varepsilon < \varepsilon_{\text{crit}}$ certifies $\|J'_z\|_2 < 1$. The certificate is one-sided: $\|J'_z\|_2 \rightarrow 1$ need not drive $\min_i |1 - \lambda_i| \rightarrow 0$ (e.g. $M = \text{diag}(-s, 0)$ has $\|M\|_2 = s$ yet $\|(I - M)^{-1}\|_2 = 1$). When J'_z is normal with dominant real-positive eigenvalue $\lambda^* = \|J'_z\|_2$ (the Perron mode for symmetric $\hat{A}, W$), $\min_i |1 - \lambda_i| = 1 - \lambda^* = \|W\|_2(\varepsilon_{\text{crit}} - \varepsilon)$ yields the $\Omega(1/(\varepsilon_{\text{crit}} - \varepsilon))$ rate; a negative or complex dominant eigenvalue keeps $\min_i |1 - \lambda_i|$ away from 0 (the same $\text{diag}(-s, 0)$), so $\varepsilon_{\text{crit}}$ only lower-bounds the threshold, slack controlled by η (Observation 1). For $\varepsilon \geq \varepsilon_{\text{crit}}$ Banach contraction fails and the iteration may converge elsewhere, oscillate, or diverge. $\square$

$\varepsilon_{\text{crit}}$ is a closed-form sufficient bound that S_c tightens $7\text{--}14\times$ over the unconstrained $\sqrt{r}$ bound (section V-A). The Neumann bound uses κ , not $\rho(J_z)$; the gap is the pseudospectral index η , bounded by W alone:

Observation 1 (Graph-Independent Nonnormality Bound, all-active case). *For $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ with symmetric $\hat{A}$ and all activations positive ($\phi' \equiv 1$): $\eta \leq \kappa(V_W)$ where $W = V_W D_W V_W^{-1}$ and $\kappa(V_W) = \|V_W\|_2 \|V_W^{-1}\|_2$, independent of $\hat{A}$.*

Proof. $J_z = \hat{A} \otimes W$ has Kronecker eigenvalues $\lambda_i(\hat{A})\lambda_j(W)$ [32]; symmetric $\hat{A}$ gives $\kappa(U_{\hat{A}}) = 1$, so the joint eigenvector matrix has condition number $\kappa(V_W)$, and the diagonalisable bound $\|(I - M)^{-1}\|_2 \leq \kappa(V)/(1 - \rho(M))$ yields $\eta \leq \kappa(V_W)$. $\square$

Remark 1 (Empirical extension to general ReLU patterns). *The proof above requires $\phi' \equiv 1$; for general ReLU patterns, $\eta \in [1.19, 2.47]$ on our suite (section V-E), tracking $\kappa(V_W)$ closely. Graph structure does not amplify nonnormality; the moderate η traces to spectral-norm regularisation of W .*

With the resolvent's amplification characterised, we now read the three diagnostics off S_c , beginning with the single most damaging perturbation direction.

Proposition 1 (Maximally Sensitive First-Order Structural Perturbation). *The perturbation δA^* maximizing $\|\Delta z^*\|$ to first order subject to $\|\delta A\|_F \leq \varepsilon$ is $\delta A^* = \varepsilon \cdot \text{reshape}(v_1, N \times N)$, where v_1 is the leading right singular vector of S , with shift $\varepsilon \cdot \sigma_1(S)$. Restricted to the threat model's feasible set (symmetric, edge-supported; section II) the maximizer is instead the leading right singular vector of the constrained S_c (below) in the edge basis $\{b_k\}$, with shift $\varepsilon \sigma_1(S_c) \leq \varepsilon \sigma_1(S)$; the unconstrained $\text{reshape}(v_1)$ is generally infeasible.*

By the variational characterisation of σ_1 , the direction (Proposition 1) is optimal for hidden-state damage within the pseudospectral envelope of Observation 1 and Remark 1, and empirically aligns with classification damage (section V-B). The constrained sensitivity matrix $S_c \in \mathbb{R}^{N \times |E|}$ has columns $[S_c]_{:,k} = S_{:,iN+j} + S_{:,jN+i}$ paired with edge basis $b_k = (e_i e_j^\top + e_j e_i^\top)/\sqrt{2}$ ($\|b_k\|_F = 1$, so $\sigma_1(S_c)$ is consistent with $\|\delta \hat{A}\|_F = \varepsilon$). The $N^2 \rightarrow |E|$ projection P_c is the standard duplication-matrix reduction [33] on the edge-supported

symmetric subspace; the operational contribution is the matrix-free $S_c v$ in algorithm 1. The first-order envelope identifies the dangerous direction rather than a global safety margin (section V-A).

Proposition 2 (Per-Node First-Order Sensitivity Radius). *For a linear head $f(z) = Wz + b$ and a node v classified as y_v with positive margins $m_v^{(c)} = f_{y_v}(z_v^*) - f_c(z_v^*) > 0$ to every competitor $c \neq y_v$, the first-order sensitivity radius*

$$r_v = \min_{c \neq y_v} \frac{m_v^{(c)}}{\|(W_{y_v} - W_c) S_v\|_2} \quad (7)$$

preserves the classification of v for any $\|\delta \hat{A}\|_F < r_v$, where S_v are the block-rows of S at node v ; substituting $S_{c,v}$ gives the tighter constrained-threat radius $r_v^{(c)} \geq r_v$. Minimizing over all competitors is the distance to the nearest first-order boundary, so the bound holds even when the runner-up changes under perturbation; the cheaper runner-up surrogate $\hat{r}_v = m_v / (\|W_{y_v} - W_{c^}\|_2 \|S_v\|_2)$ ($c^* = \arg \max_{c \neq y_v} f_c$) can be optimistic when a non-runner-up class is more aligned.*

Proof sketch. By Thm. 1(a), $\Delta z_v^* \approx S_v \text{vec}(\delta A)$, so for each competitor $|\Delta(f_{y_v} - f_c)| \leq \|(W_{y_v} - W_c) S_v\|_2 \|\delta A\|_F$ by Cauchy-Schwarz; the prediction survives while every margin stays positive, i.e. $\|\delta A\|_F < m_v^{(c)} / \|(W_{y_v} - W_c) S_v\|_2$ for all c , whose minimum is r_v . Intuitively, r_v is the smallest margin divided by two amplification factors (decision-boundary sensitivity to hidden-state change, and equilibrium sensitivity to structure); high-margin, low-sensitivity nodes get large radii.

Remark 2 (Sensitivity threshold semantics). r_v is a first-order sensitivity threshold, not a probabilistic certificate: it indicates the perturbation scale at which the linearization predicts a classification change for node v , and is locally tight for small ε but can be violated at larger magnitudes where higher-order terms dominate. The two approaches are complementary: randomized smoothing [7] certifies which nodes are guaranteed safe, while AEGIS differentiates which edges and nodes are most sensitive and in which direction (section V-B). Empirically, breach rate is 0% at $\varepsilon = 0.01$ and rises with ε as expected for first-order approximations (section V-B).

A. Continuous-to-Discrete Transfer

Proposition 3 (Continuous-to-Discrete Ranking Transfer). *Under (A1)–(A3), let $w_k = [\hat{A}]_{ij}$ denote the normalized weight of edge $k = (i, j) \in E$, $v_k = \|[S_c]_{:,k}\|_2$ the continuous vulnerability score, and $d_k = \|z^*(A) - z^*(A \setminus k)\|_2$ the discrete damage from fixed-normalization removal of edge k . Assume all single-edge removals are subcritical: $\sqrt{2} \max_k w_k < \varepsilon_{\text{crit}}$.*

(a) First-order bridge. The discrete damage satisfies

$$d_k = w_k \cdot v_k + R_k, \quad |R_k| \leq \frac{L_J}{2(1 - \kappa)^2} \cdot w_k^2, \quad (8)$$

where $L_J \leq \|W\|_2^2 \|z^\|$ is the second-order curvature constant of $z^*(A)$, finite at the fixed point.*

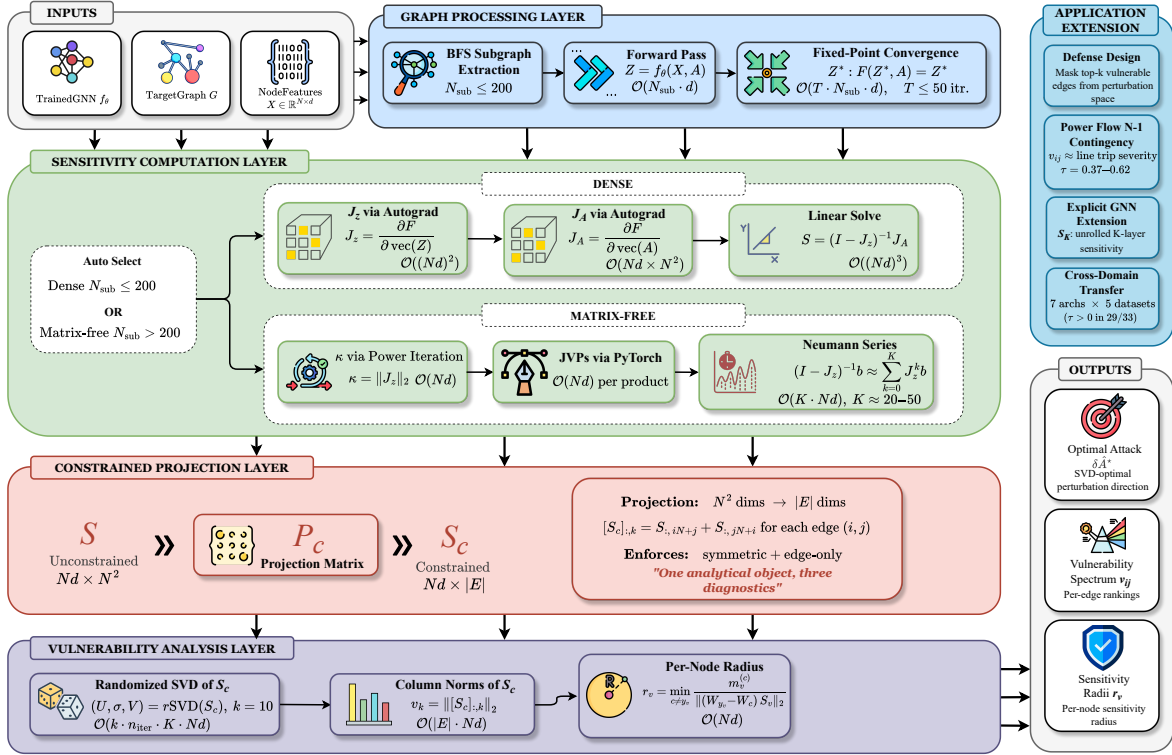


Fig. 1: The four-stage AEGIS pipeline: (1) graph processing (BFS subgraph or full-graph forward / fixed-point pass); (2) sensitivity computation (dense Jacobian for $N \leq 200$, matrix-free Neumann/JVP otherwise); (3) constrained projection $N^2 \rightarrow |E|$ via P_c ; (4) vulnerability analysis (randomised SVD, S_c column norms, per-node radii). The matrix-free path queries S_c via JVPs/VJPs without materialising it, scaling to $N=7,650$.

(b) Ranking preservation (pairwise-sufficient). For two edges k_1, k_2 with $v_{k_1} > v_{k_2}$ and $w_{k_1} \geq w_{k_2}$, we have $d_{k_1} > d_{k_2}$ whenever

$$w_{k_1} < \frac{v_{k_1} - v_{k_2}}{L_J / (1 - \kappa)^2}. \quad (9)$$

Under uniform weights ($w_k = w$ for all k), $v_{k_1} > v_{k_2}$ implies $d_{k_1} > d_{k_2}$ whenever $w < \min_{k_1 \neq k_2} (v_{k_1} - v_{k_2}) \cdot (1 - \kappa)^2 / L_J$.

Proof. (a). We model deletion as *fixed-normalization* masking, $[\delta \hat{A}]_{ij} = [\delta \hat{A}]_{ji} = -w_k$ with the degree matrix D held fixed, matching the continuous edge-weight perturbation, so by the S_c construction $\|\Delta z^*\| \approx w_k v_k$. (recompute-normalization adds an $\mathcal{O}(d_i^{-1})$ incident-edge rescaling, negligible off low-degree nodes; agreement in section V-A.) For the remainder, the IFT gives $\partial z^* / \partial \text{vec}(A) = (I - J_z)^{-1} J_A$, and the dominant curvature term $(I - J_z)^{-1} (\partial J_z / \partial \text{vec}(A)) (I - J_z)^{-1} J_A$ carries two resolvents $(1 - \kappa)^{-1}$, two factors $\|W\|_2$, and the equilibrium norm $\|z^*\|$ (since $\partial J_z / \partial \text{vec}(A) \propto W$ and $J_A \propto z^* W^\top$), so $L_J \leq \|W\|_2^2 \|z^*\|$, finite since $\|z^*\| \leq \|X_{\text{proj}}\| / (1 - \kappa)$ (the κ -contraction of F at the fixed point, A3), giving $\|\partial^2 z^* / \partial \text{vec}(A)^2\|_2 \leq (1 - \kappa)^{-2} L_J$. The path meets finitely many ReLU regions (simultaneous crossings non-generic, measure zero); summing the second-order remainder over these regions, with the conservative IFT [31] on each, gives $|R_k| \leq L_J w_k^2 / (2(1 - \kappa)^2)$. **(b).** From (a), $d_{k_1} - d_{k_2} \geq w_{k_1} v_{k_1} - w_{k_2} v_{k_2} - C(w_{k_1}^2 + w_{k_2}^2)$ with $C = L_J / (2(1 - \kappa)^2)$; under $v_{k_1} > v_{k_2}$, $w_{k_1} \geq w_{k_2}$, the first term is positive and (9)

ensures the first-order gap dominates. $\square$

Empirically $|R_k|$ is $2-10\times$ smaller than $L_J w_k^2$, and the constrained projection S_c tightens the bound from 0.31 to 1.00 (section V-A). The edge-weighted score $w_k v_k$ predicted by (a) reaches $\tau = +0.996$ against brute-force N-1 on full-graph Amazon Photo (10 seeds, section V-F), corroborating the first-order bridge at scale.

B. Generalization to Explicit GNNs

Proposition 4 (Structural Sensitivity for K -Layer GNNs). Let $Z_K = g_K \circ \dots \circ g_1(X, A)$ be a K -layer GNN where each layer g_l is differentiable with respect to both its input and A . Define the unrolled sensitivity matrix

$$S_K = \frac{\partial \text{vec}(Z_K)}{\partial \text{vec}(A)} = \sum_{l=1}^K \left(\prod_{k=l+1}^K J_z^{(k)} \right) J_A^{(l)}, \quad (10)$$

where $J_z^{(k)} = \partial g_k / \partial Z_{k-1}$ and $J_A^{(l)} = \partial g_l / \partial \text{vec}(A)$. Then:

(a) The first-order shift bound $\|\Delta Z_K\|_F \leq \sigma_1(S_K) \cdot \|\delta A\|_F + \mathcal{O}(\|\delta A\|^2)$ holds, with

$$\sigma_1(S_K) \leq \sum_{l=1}^K \left(\prod_{k=l+1}^K \|J_z^{(k)}\|_2 \right) \|J_A^{(l)}\|_2. \quad (11)$$

(b) The constrained construction S_c and per-node radius r_v (Proposition 2) apply identically with S_K in place of S .

For weight-tied models, $\sigma_1(S_K) \leq \|J_A\|_2 (1 - \kappa^K) / (1 - \kappa)$ converges to the IGNN bound as $K \rightarrow \infty$. Explicit GNNs inherit

Algorithm 1 AEGIS matrix-free vulnerability analysis. S_c is queried but never formed; the dense fallback ($N \leq 200$) builds S_c and runs an exact SVD.

Require: F , fixed point z^* , $\hat{A}$, edge set E , budget ε , rank k , Neumann tolerance tol ; (opt.) head W , margins $m_v^{(c)}$.

Ensure: $\sigma_1(S_c)$, $\delta\hat{A}^*$, $\{v_{ij}\}$, $\{r_v\}$, $\varepsilon_{\text{crit}}$.

- 1: $\kappa \leftarrow \|J_z\|_2$; $K \leftarrow \lceil \log(1/\text{tol}) / \log(1/\kappa) \rceil$
- 2: $S_c v := (\sum_{j=0}^K J_z^j) J_A P_c v$ {matrix-free}
- 3: $(\sigma_1, v_1) \leftarrow \text{rSVD}(S_c, k)$ [34]
- 4: $\delta\hat{A}^* \leftarrow \varepsilon \text{sym}(P_c v_1) / \|\text{sym}(P_c v_1)\|_2$
- 5: $v_{ij} \leftarrow \|S_c e_{ij}\|_2$ for each $(i, j) \in E$
- 6: $r_v \leftarrow \min_{c \neq y_v} m_v^{(c)} / \|(W_{y_v} - W_c) S_{c,v}\|_2$ {head only}
- 7: $\varepsilon_{\text{crit}} \leftarrow (1 - \kappa) / \|W\|_2$ {Thm. 1}
- 8: **return** $(\sigma_1, \delta\hat{A}^*, \{v_{ij}\}, \{r_v\}, \varepsilon_{\text{crit}})$

the full computational pipeline with empirical tightness 0.99–1.02 across the architecture suite (section V-F); the closed-form $\varepsilon_{\text{crit}}$ is available only for the contractive subclass.

IV. AEGIS CONSTRUCTION

The pipeline makes the theory of section III practical: it queries S_c matrix-free, never materialising $S \in \mathbb{R}^{Nd \times N^2}$ or forming the resolvent. One S_c computation returns the per-edge spectrum $\{v_{ij}\}$, the SVD-optimal direction $\delta\hat{A}^*$, and per-node radii $\{r_v\}$, plus $\varepsilon_{\text{crit}}$ and the regime characterisation of Theorem 1 for IGNN-class operators. A dense path ($N \leq 200$) computes an exact SVD; a matrix-free path ($N > 200$, validated to $N=7,650$) uses JVP/VJP queries and randomized SVD. Figure 1 illustrates the four stages; algorithm 1 summarises the matrix-free routing.

Matrix-free operator. The key to scaling is to never build S_c : we only apply it to vectors, which needs $O(Nd)$ memory rather than the $O((Nd)^2)$ a dense S_c would cost. Concretely, $S_c v = (I - J_z)^{-1} J_A P_c v$ is evaluated via the truncated Neumann series $\sum_{k=0}^K J_z^k b$ with depth K adaptive in κ (early stop at $\|J_z^k b\| < 10^{-6} \|b\|$); each Neumann term and $J_A \text{vec}(\delta A)$ are forward-mode JVPs in $O(Nd)$ time. P_c is applied implicitly, so neither S_c nor S is formed; the dense path ($N \leq 200$) materialises J_z, J_A and applies a direct linear solve. The rSVD [34] ($k=p=10, n_{\text{iter}}=2$) returns (u_1, σ_1, v_1) for the direction $\delta\hat{A}^* = \varepsilon \cdot \text{sym}(P_c v_1)$ (Proposition 1), where $\text{sym}(M) = (M + M^T)/2$ keeps $\delta\hat{A}^*$ symmetric (section II); column norms give $\{v_{ij}\}$, and per-node radii follow Proposition 2 via the node-restricted RMATVEC. The well-separated singular spectrum (gap $(\sigma_1 - \sigma_2)/\sigma_1 = 0.39\text{--}0.50$; fig. 2) is why one rSVD query matches iterative attackers (section V-B).

Cost. Each $S_c v$ is $O(K \cdot Nd)$ time and $O(Nd)$ memory ($K \in [20, 50]$ for $\kappa < 0.8$); removing the $O((Nd)^3)$ solve lifts the previous $N \approx 300$ ceiling (section V-D).

V. EXPERIMENTS

Our evaluation tests four claims in turn: the contractivity assumptions hold on trained models (section V-A); the one-query SVD direction matches iterative attackers (section V-B); the continuous S_c ranking transfers to discrete edge removal

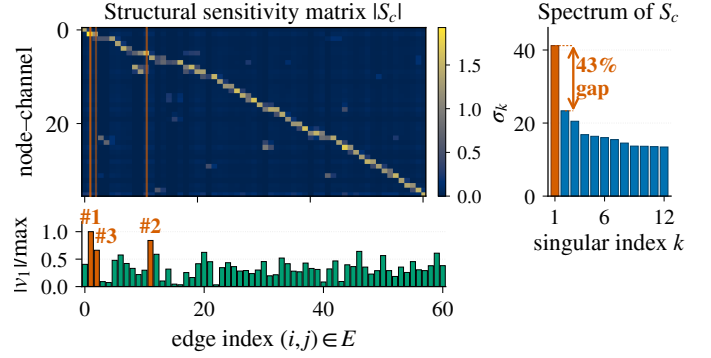


Fig. 2: $|S_c|$ on a 50-node Cora ego-graph (IGNN, $N=50, d=16, |E|=61$, seed 0). Vermilion lines: top-3 edges by $|v_1|$ (SVD-optimal direction, bar strip). Sidebar: singular spectrum, $\sigma_1=41.2$, 43% gap to σ_2 (cross-benchmark $(\sigma_1 - \sigma_2)/\sigma_1 \in [0.39, 0.50]$; section V-B).

TABLE I: Per-dataset IGNN characterisation (10 seeds). $\kappa = \|J_z\|_2$ verifies (A3); $\varepsilon_{\text{crit}} = (1 - \kappa) / \|W\|_2$ is the local critical budget. AtkAdv: AEGIS damage / random damage. Cov%: fraction of subgraph nodes certified first-order-safe at budget $\varepsilon=0.05$ (i.e. corrected radius $r_v > 0.05$).

Dataset	Test Acc	κ	$\varepsilon_{\text{crit}}$	AtkAdv	Cov%
Cora	77.5 $\pm$ 1.7	.33 $\pm$.14	.66 $\pm$.15	3.6 $\pm$.6	83 $\pm$ 10
Citeseer	66.0 $\pm$ 0.7	.59 $\pm$.02	.41 $\pm$.02	4.1 $\pm$.5	94 $\pm$ 5
Pubmed	78.9 $\pm$ 0.7	.41 $\pm$.11	.59 $\pm$.11	3.2 $\pm$.4	76 $\pm$ 18
Amazon	94.8 $\pm$ 0.3	.14 $\pm$.02	.86 $\pm$.02	3.8 $\pm$.2	86 $\pm$ 8
WikiCS	77.9 $\pm$ 0.4	.34 $\pm$.04	.66 $\pm$.04	3.8 $\pm$.4	78 $\pm$ 4

across architectures (section V-F); and the same object audits a deployed fraud detector (section VI).

Setup. We evaluate AEGIS on 6 datasets across 4 domains with 10 random seeds each (PyTorch [35]); reported $\pm$ are standard deviations over seeds. IGNN [22] uses $F(Z) = \text{ReLU}(\hat{A}ZW^T + U(X))$ with $W \in \mathbb{R}^{64 \times 64}$ spectral-normalised [23] and Adam (lr 0.01). The spectral-norm constraint costs $\sim 6\%$ Cora accuracy. Datasets: Cora, Citeseer, Pubmed [36], Amazon Photo [37], WikiCS [38], Amazon Fraud [39]. Baselines: smoothing (Gaussian $\sigma=0.01$), Mettack (Meta-Self GCN surrogate). Coverage is not uniform across studies: all seven architectures appear only in table IV and fig. 7 (39 cells), while the other studies use IGNN.

A. Cross-Domain Adversarial Analysis

Two prerequisites must hold: the trained models sit in the contractive regime the theory assumes, and AEGIS’s perturbations beat chance. Table I confirms both, verifying (A3) on every benchmark ($\kappa < 1$) and reporting per-dataset $\varepsilon_{\text{crit}}$ alongside a $3.2\text{--}4.1\times$ attack advantage over random. Unless otherwise stated, IGNN experiments in section V-A–section V-E use 50-node BFS ego-subgraphs centred on the highest-degree node and report *local vulnerability*; graph-level assessment uses the matrix-free pipeline of section V-D. All bounds use the operator-norm $\kappa = \|J_z\|_2$.

AEGIS’s shift bounds are first-order; predicted and actual equilibrium shifts agree within 1% at the small budgets that matter for early warning, loosening to at most 39% on the citation graphs (within 7% on WikiCS and Amazon) even at $\varepsilon=0.20$. This envelope ratio is measured on the equilibrium

TABLE II: Equilibrium damage at $\varepsilon=0.10$ across attack methods (IGNN, 50-node subgraph, 10 seeds). SVD: AEGIS. Cls-PGD: classification-loss PGD. Shift-PGD: IFT-gradient PGD (solver validation, not an independent baseline).

Dataset	SVD	Cls-PGD	Shift-PGD	Random
Cora	$3.70 \pm .79$	$2.51 \pm .48$	$3.05 \pm .70$	$0.97 \pm .25$
Citeseer	$4.63 \pm .71$	$2.97 \pm .42$	$3.35 \pm .57$	$1.01 \pm .15$
WikiCS	$2.10 \pm .22$	$0.67 \pm .11$	$1.93 \pm .20$	$0.53 \pm .09$

shift $\|\Delta z^*\|_F$, distinct from the prediction-flip (breach) rate of fig. 4.

Heuristic and surrogate baselines. AEGIS beats degree-proportional, edge-betweenness, and spectral rankings at $\varepsilon=0.01$ (+6–148% AtkAdv, Wilcoxon $p<0.001$) and inflicts 3–10 $\times$ more equilibrium damage than Mettack [4] at the early-warning regime $k \in \{1, \dots, 5\}$ (149/150 paired wins, $p<10^{-43}$); GR-BCD/PR-BCD [5] and the discrete-removal curves are in table III and fig. 3.

B. Four-Quadrant Attack Comparison

Radius vs. smoothing. Randomized smoothing [7] gives constant-per-node base radii (0.123 vs. AEGIS’s 0.078 at $\sigma=0.05$), and localized smoothing [9] needs 10^3 – 10^4 MC/node; both lack edge structure. AEGIS radii are instead structurally informative ($r_v \approx 0.10$ in dense regions, 0.01 at boundaries), a complementary point on the certificate-versus-diagnostic frontier.

Table II evaluates AEGIS across the four quadrants of (gradient-based, gradient-free) $\times$ (equilibrium-shift, classification-loss). The SVD direction maximises first-order equilibrium shift by construction (Proposition 1); what matters is non-circularity and cost. A transfer direction from a separately trained surrogate (zero shared gradients) recovers 99% of the one-query damage ($\cos=0.99$), versus $44 \pm 4\%$ for a 512-query black-box search: model-intrinsic, not a gradient artifact. Per query it delivers 74–156 $\times$ the damage of a 50-step PGD [40]; classification-loss PGD inflicts 15–70% less at 50 $\times$ the cost, IFT-gradient PGD (solver validation, not independent) 72–92%, and prediction *flips* stay 0–1.8% for all methods. AEGIS is a one-query *diagnostic*, not a peak attacker: budget-heavy attackers (GR-BCD, table III) exceed it at larger budgets.

Classification-gradient edges differ. Per-edge classification gradients anti-correlate with discrete ground truth on Cora ($\tau = -0.20 \pm 0.05$ vs. AEGIS’s $+0.32 \pm 0.04$) and stay near zero elsewhere; AEGIS achieves 2–6 $\times$ higher P@10, confirming that equilibrium and classification sensitivities are distinct vulnerability surfaces.

Per-edge finite differences reproduce the S_c column-norm ranking ($\tau=0.999$); iterative S_c recomputation closes ≤ 3 pp of the static-vs-greedy gap at $k \times$ compute, so the static ranking is the headline.

Breach rates. Every breached node satisfies $\varepsilon > r_v$ across all datasets and budgets, validating r_v as a reliable first-order screening threshold (Remark 2). Figure 4 reports prediction-flip fractions across ε (bands are 95% CIs): Citeseer and WikiCS stay below 2% throughout; Cora reaches 7.6% (CI [2.2, 13.0])

TABLE III: AEGIS vs. external baselines (10 seeds). GR-BCD [5] is the BCD-family scalable structural attacker (PR-BCD is the same paper’s larger-budget variant, see text); AGNNCert [11] is a sound IBP certifier (semantics differ from AEGIS r_v).

Baseline	Setting	AEGIS	Baseline	τ
GR-BCD [5]	Pubmed $k=10$	0.356	0.350	+0.69
GR-BCD [5]	Cora $k=5$	0.643	1.207	+0.16
AGNNCert [†] [11]	Cora med. r_v	0.163	1.414	+0.08

[†]AGNNCert is a *sound* IBP certificate; AEGIS r_v is a first-order sensitivity threshold (Remark 2). Per-dataset mean tightness ratio $r_{\text{cert}}/r_v \in [4.4, 15.0]$ (per-cell across 30 seeds [2.1, 39.0] $\times$); per-seed Kendall $\tau \in [-0.11, 0.24]$. **Decision rule:** if AGNNCert certifies node v safe at budget ρ , trust it; if $r_v < \rho$ and AGNNCert does *not* certify, v is first-order unsafe and warrants attention. The methods are complementary in this prescribed sense, not interchangeable.

at $\varepsilon=0.20$; Pubmed is the right-skewed outlier (median 7.8%, mean 10.3% at $\varepsilon=0.10$; 27.4%, CI [12.3, 42.5], at $\varepsilon=0.20$).

C. Comparison with Prior Structural Baselines

Table III reports head-to-head numbers against GR-BCD and PR-BCD [5] and AGNNCert [11]. AEGIS is a label-free one-query proxy; the question is how much of each gold-standard signal it retains.

GR-BCD [5] aligns with S_c on Pubmed ($\tau=+0.69$) but decorrelates on Cora ($\tau=+0.16$); the budgets shown ($k \leq 10$) are AEGIS’s early-warning regime, where its one-pass diagnostic is competitive, while GR-BCD/PR-BCD dominate raw damage at the high budgets they target (table III). AGNNCert [11] and r_v correlate weakly (near-zero per-seed τ , table III), precisely because they flag *different* nodes — they are complementary, not redundant: r_v is a sound first-order screen (no node breaches below it, section V-B), 4.4–15.0 $\times$ tighter than the certificate.

D. Phase Transition and Scalability

Phase transition. Figure 6 sweeps $\kappa_{\max} \in [0.30, 0.99]$ on Cora (10 seeds). Even when the spectral-normalisation cap is pushed to 0.99, driving the theoretical $\varepsilon_{\text{crit}}$ down 230 $\times$, the trained ReLU pattern keeps $\rho(J_z) \leq 0.42$, so the resolvent grows only 1.17 $\rightarrow$ 1.80. This is a 2–4 $\times$ spectral-radius margin to criticality ($\rho=1$), the operative quantity behind $\varepsilon_{\text{crit}}$, stable across the sweep. Training converges throughout, and the matrix-free pipeline reaches its 50-step fixed-point budget at $\kappa_{\max} \geq 0.85$.

Scalability and matrix-free self-consistency. Figure 5 compares dense and matrix-free pipelines. The matrix-free $S_c v$ implements the dense S_c up to Neumann truncation; σ_1 agrees within 0.03% at $N=200$ (per-edge $\tau=0.999$), and the analytical truncation residual $\kappa^{200} \in [10^{-105}, 10^{-48}]$ across the suite (consistent with the trained-IGNN range $\kappa \in [0.14, 0.59]$ of table I). For larger N the rSVD error [34] is bounded by the spectral gap; Pubmed ($N=19,717$) exceeds 24 GB.

E. Ablations and Defense

Subgraph size. Varying $N \in \{30, 50, 100, 200\}$ on Cora (10 seeds): tightness is stable at 1.01–1.02 for $N \leq 100$ and degrades to 1.031 at $N=200$; we use $N=50$ as default (66 $\times$ faster).

Full-graph scale. A 50-node BFS covers only $\sim 1.8\%$ of a citation graph’s edges (Cora $\tau=0.16$), so at this scale we

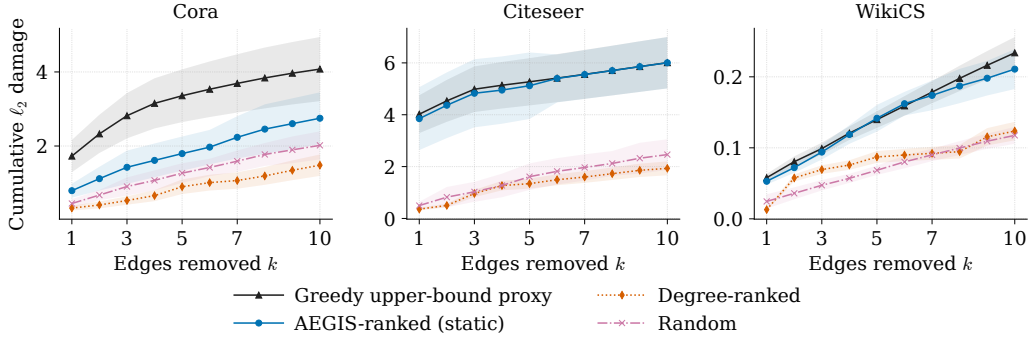


Fig. 3: Cumulative damage from sequential discrete edge removal vs. k (IGNN, 50-node subgraph, 10 seeds, ± 1 sd). AEGIS matches the ground-truth-aware Greedy attacker on Citeseer/WikiCS and recovers 54–67% of Greedy’s damage on Cora at $k=5-10$ without any access to discrete ground truth; degree-ranked stays below random on Cora at every k .

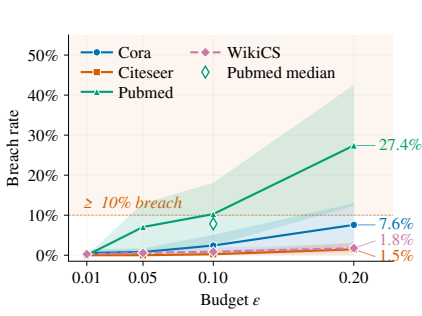


Fig. 4: Breach rate (% of nodes flipped under S_c -optimal perturbation) vs. ε (10 seeds; mean, 95% CI bands).

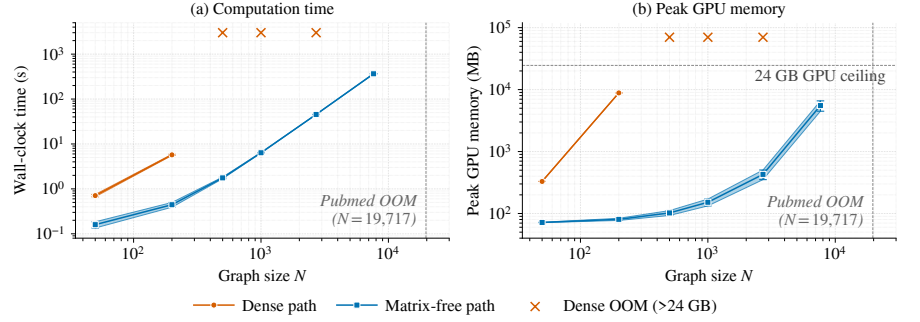


Fig. 5: Dense vs. matrix-free AEGIS pipeline (RTX 4090). Dense scales as $O((Nd)^3)$, exceeding 24 GB beyond $N=200$; matrix-free is roughly $O(N \log N)$ in time, sub-linear in memory, reaching Amazon Photo ($N=7,650$) at 365 s/5.5 GB.

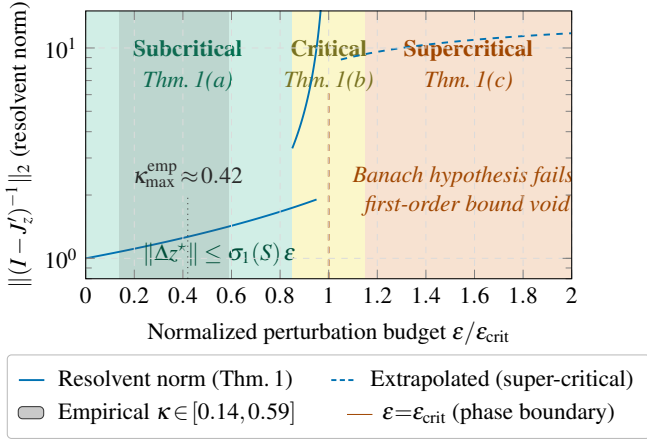


Fig. 6: Three-regime characterisation of Theorem 1. The empirical κ band (grey) saturates well below the spectral-normalisation ceiling, so the empirical resolvent growth is far milder than the worst-case $1/(1-\kappa)$ divergence at $\varepsilon \rightarrow \varepsilon_{\text{crit}}$.

run AEGIS on the full graph. There its edge advantage over degree-proportional ranking *amplifies* to $9.82\times$ (Citeseer) and $3.25\times$ (Cora) at $k=10$, against $\approx 1.1\times$ on subgraphs.

Hyperparameters. Tightness is stable across $d \in \{16, 32, 64, 128\}$; the spectral-norm ceiling $c \in \{0.5, 0.9\}$ traces the accuracy–robustness frontier ($r_v=0.147@72.1\%$ vs. $0.089@80.6\%$).

Edge protection and the delocalization of vulnerability.

On the *full* graph (Cora, $N=2,708$, 10 seeds), masking the top- k edges by v_{ij} cuts $\sigma_1(S_c)$ damage by $2.4 \pm 1.8\%$ at $k=5$ and $4.6 \pm 2.9\%$ at $k=10$ — $12\text{--}46\times$ over random masking, but far below the $42\text{--}61\%$ a 50-node subgraph reports (where $k=5$ is already 10% of its edges). The reason is structural: the leading attack mode is *delocalized* (participation ratio 41–89 edges), so removing a handful excises a negligible slice; reaching subgraph-scale reduction needs 2–8% of edges masked, and an adaptive recomputation erodes the gain a further 1–2 pp. AEGIS thus *localizes* risk for attribution and triage rather than yielding a few-edge defense, the failure mode sober-look critiques [16] flag.

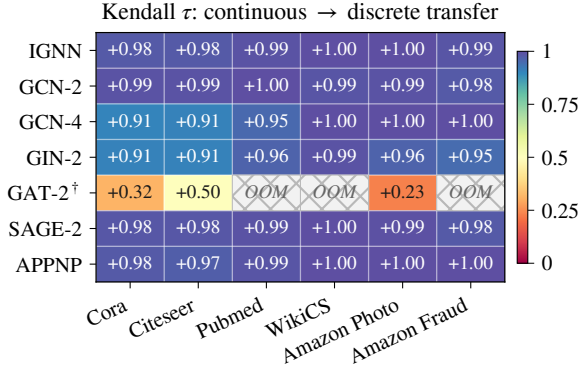
F. Extension to Explicit GNNs

The construction is not tied to implicit models. For K -layer explicit GNNs (GCN, SAGE, GIN, APPNP, GAT[†] [41], [42], [43], [44], [45]) we form $S_K = \partial \text{vec}(Z_K) / \partial \text{vec}(A)$ via finite differences and build S_c from S_K identically (Proposition 4). Table IV reports the per-architecture characterisation on Cora: IGNN carries Theorem 1; the explicit models receive the computational tool without the regime characterisation.

GAT[†] scope. Standard GAT uses A as a binary mask, so $\partial Z / \partial A_{ij} = 0$ on existing edges and S_c is undefined; our GAT[†] modulates attention by the edge weight ($h'_i = \sum_j \hat{A}_{ij} \alpha_{ij} W h_j$), restoring differentiability and yielding tightness 0.99, AtkAdv $4.4\times$, $\tau = +0.36$. Binary-mask architectures (hard attention,

TABLE IV: S_c framework on 7 GNN architectures (Cora, 10 seeds). Tight.: actual/predicted shift ratio. AtkAdv: AEGIS/random damage. τ : Kendall of the unweighted v_{ij} ranking vs. single-edge removal; the edge-weighted $A_{ij}v_{ij}$ ranking of Proposition 3 (fig. 7) is the higher headline score. Acc: full-graph test accuracy.

Model	Tight.	AtkAdv	τ	Acc
IGNN (eq.)	1.01 $\pm$.00	7.6 $\pm$ 0.5 $\times$	+ .32 $\pm$.04	77.5 $\pm$ 1.7%
GCN-2	1.00 $\pm$.00	3.6 $\pm$ 0.6 $\times$	−.04 $\pm$.03	78.9 $\pm$ 0.9%
GCN-4	1.02 $\pm$.00	3.4 $\pm$ 0.4 $\times$	+ .49 $\pm$.02	78.3 $\pm$ 1.8%
GIN-2	1.01 $\pm$.00	2.6 $\pm$ 0.1 $\times$	+ .33 $\pm$.03	76.6 $\pm$ 1.9%
GAT [†]	1.01 $\pm$.01	2.1 $\pm$ 0.1 $\times$	+ .54 $\pm$.06	77.8 $\pm$ 1.2%
SAGE-2	1.00 $\pm$.00	1.9 $\pm$ 0.2 $\times$	+ .22 $\pm$.10	77.5 $\pm$ 1.5%
APNP	1.02 $\pm$.00	3.4 $\pm$ 0.3 $\times$	+ .35 $\pm$.01	82.2$\pm$0.6%



[†] GAT-2: 2-layer attention; OOM on Pubmed, WikiCS, Amazon Fraud.

Fig. 7: Continuous-to-discrete transfer τ (10-seed means, 390 runs; per-cell sd 0.0003–0.041), ranking edges by the edge-weighted $A_{ij}v_{ij}$ of Proposition 3. All 39/39 cells positive (median +0.99); cross-hatch: GPU OOM. GAT-2[†] is lower under weighting (it transfers better unweighted; see text).

max/min aggregation, GATv2 [46]-style dynamic-attention masking) fall outside the framework.

Robust backbones. Under the $\sigma_1(W) \leq 0.9$ cap, RobustGCN-lite [13] and GNNGuard-lite [14] match or exceed IGNN; the cap is a precondition (without it $\kappa > 1$ voids Theorem 1).

Cross-dataset transfer. We rank edges by the score Proposition 3 predicts, the edge-weighted $A_{ij}v_{ij}$. Figure 7 reports its τ over 6 datasets and 7 architectures (390 runs): *all* 39/39 cells positive, median $\tau = +0.99$ (sign test $p < 10^{-5}$), vs. 34/39 for unweighted v_{ij} . It beats a pure edge-weight baseline by $\Delta\tau \approx +0.25$, rising to $+0.65$ on the fraud graph where uniform weights are uninformative, so v_{ij} adds rank information beyond the weight. Weighting matters most at full-graph scale: on Amazon Photo ($N=7,650$; full-graph $\kappa \approx 1.00$, marginally non-contractive, so empirical) it reaches $\tau = +0.996 \pm 0.002$ (stratified top- v_{ij} N-1). GAT-2[†] is the exception: attention reweights edges, so the unweighted v_{ij} transfers better there (+0.47 vs. +0.35).

Proposition 3’s sufficient pairwise condition holds for 47–62% of edge pairs, so the global τ is empirical, not implied. Where unweighted v_{ij} is uninformative (GCN-2’s near-uniform 2-hop shifts), the weighted ranking recovers the order via the edge weight ($-0.28 \rightarrow +0.99$, Citeseer).

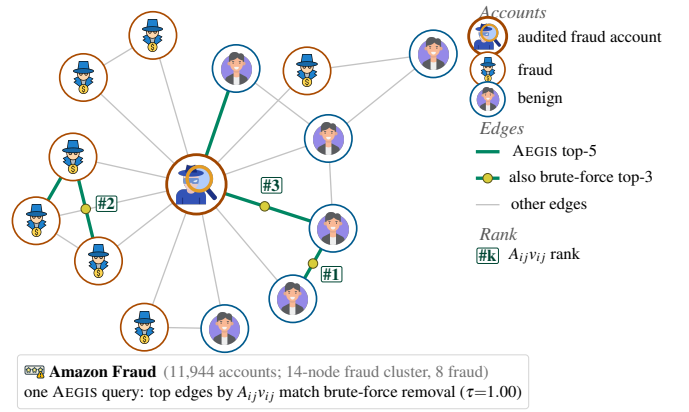


Fig. 8: AEGIS auditing an IGNN fraud detector on an Amazon Fraud cluster [39]. Icons mark fraud / benign accounts; the audited account sits at the hub. Green edges are the top-5 adversarial fault lines by the edge-weighted score $A_{ij}v_{ij}$ of Proposition 3; yellow dots mark the three also in the brute-force single-edge-removal damage top-3. The one-query ranking reproduces the brute-force ranking ($\tau=1.0$ on this cluster), at one query’s cost rather than $|E|$ reconvergences.

VI. CASE STUDY: AUDITING A FRAUD DETECTOR

GNN fraud detectors are a prime adversarial target: a flagged account can try to evade detection by perturbing a few of its behavioural-similarity edges [1], [39]. AEGIS audits such a detector directly — no labels, no retraining — reading from one S_c query which edges most threaten its predictions.

Figure 8 shows the audit: the trained IGNN flags the central account, and the edge-weighted ranking $A_{ij}v_{ij}$ pinpoints the fault lines, reproducing the brute-force single-edge-removal damage ranking from a single S_c query.

This auditing claim is what the paper validates at scale, and *non-circularly*. The edge-weighted ranking transfers across the full fraud graph (fig. 7), precisely where uniform edge weights are uninformative so v_{ij} adds the most rank information ($\Delta\tau$ up to $+0.65$ over the weight baseline). The SVD direction transfers from a *separately trained* surrogate (zero shared gradients, $\cos=0.99$; section V-B), so v_{ij} reflects model-intrinsic vulnerability rather than a gradient artifact; AEGIS out-damages discrete attackers (Mettack, GR-BCD) on their own metric in the early-warning regime (sections V-A and V-C); and the per-edge v_{ij} scores prioritise which connections to monitor, though adversarial vulnerability itself proves delocalized at scale (section V-E). Unlike a certifier, S_c names *which* edges and *in which direction* (Remark 2), turning a one-query audit into an actionable defense.

VII. RELATED WORK

We position AEGIS against three threads; each supplies at most one of the three diagnostics AEGIS produces from one S_c object via a single matrix-free pipeline.

Structural attacks. Nettack [1], Mettack [4], GR-BCD/PR-BCD [5] and variants [6], [47] optimise surrogate gradients to flip predictions but provide no analytic maximally sensitive first-order direction and no per-node radius. Sober-look studies [16], [17] flag evaluation pitfalls; we follow their adaptive-attack

discipline in section V-E (paired Wilcoxon against random masking; an adaptive-attacker column in defense ablation).

Certified robustness and defense. Convex relaxations [12], smoothing [7], [8], [9], collective certificates [10], IBP [11], and weight-norm robustness bounds [48] certify per-node robustness but do not identify vulnerability-driving edges. Empirical defenses [13], [15], [14] harden models without surfacing per-edge sensitivity. r_v is a first-order threshold, not a sound certificate (Remark 2); the methods are complementary under the prescribed decision rule of table III.

Implicit networks and influence functions. DEQs [20], IGNN [22], monotone equilibria [49], and well-posedness analyses [29] address *input* sensitivity $\partial z^*/\partial x$; AEGIS targets *structural* sensitivity $\partial Z/\partial A$ via the resolvent $(I - J_z)^{-1} J_A$ and the unrolled Jacobian S_K . The construction descends from the Koh–Liang influence-function lineage [18] and implicit-differentiation hyperparameter optimisation [27], [19], but those apply IFT to loss perturbations; AEGIS applies it to a *structural-parameter* perturbation, projecting onto the edge subspace via P_c for the SVD-optimal direction in one matrix-free pass. GNN explainers [50] produce per-edge importance, the closest analogue to our per-edge ranking, but for prediction *fidelity* (which subgraph rationalises an output) rather than adversarial vulnerability; classical matrix perturbation theory [32], [26] supplies the pseudospectral toolkit. Curvature and over-squashing analyses [51], [52] and GNN stability bounds [53], [54] likewise tie edge structure to GNN sensitivity, but in aggregate and for expressivity or information flow, whereas AEGIS localises adversarial vulnerability per edge and per node.

VIII. CONCLUSION

AEGIS maps a GNN’s *adversarial fault lines*, producing three diagnostics (per-edge rankings, the SVD-optimal attack direction, and per-node first-order sensitivity radii) from one analytical object, the constrained sensitivity matrix S_c . The construction is matrix-free and applies to any GNN with continuous edge-weight-modulated message passing; for contractive implicit models it adds a regime characterisation (Theorem 1) that holds as a conservative safety boundary.

Limitations. (i) Radii r_v are first-order thresholds, not probabilistic certificates (Remark 2); standard GAT and binary-mask architectures fall outside the framework (section V-F). (ii) Full-graph matrix-free analysis is the recommended graph-level path (section V-F). (iii) Opting into the formal $\varepsilon_{\text{crit}}$ track accepts a $\sim 6\%$ accuracy cost, on par with certified-robustness trade-offs. (iv) Insertion attacks (Nettack-class) are deliberately scoped out; the S_c construction extends once the edge basis is enlarged to $\bar{E} \supseteq E$ (section II). (v) AEGIS audits a trained model’s vulnerability, not ground-truth physics: a contractive equilibrium surrogate always converges to a fixed point, so it cannot represent voltage-collapse contingencies (loss of fixed-point existence) and complements rather than replaces domain analyses such as power-flow contingency screening [55], [3], [56].

Disclosure protocol. (i) A 90-day coordinated-notification window before public release, to the affected benchmark and attacker-tool maintainers. (ii) Attack-direction code (Algorithm 1 steps 3–4) gated behind institutional-affiliation review mediated by an institutional research-ethics office (named in the camera-ready). (iii) The diagnostic-only path (r_v and v_{ij} scores; no SVD reconstruction) released unconditionally: a score *rank*s which edges are sensitive but is not itself an attack; the damaging perturbation δA^* (Proposition 1) comes only from the gated reconstruction (steps 3–4).

REFERENCES

- [1] D. Zügner, A. Akbarnejad, and S. Günnemann, “Adversarial attacks on neural networks for graph data,” in *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2018, pp. 2847–2856.
- [2] H. Dai, H. Li, T. Tian, X. Huang, L. Wang, J. Zhu, and L. Song, “Adversarial attack on graph structured data,” in *International Conference on Machine Learning*, 2018, pp. 1115–1124.
- [3] A. M. Nakiganda, C. Weeraddana, N. Popli, and T. van der Hagen, “Graph neural networks for fast contingency analysis of power systems,” *arXiv preprint arXiv:2310.04213*, 2023.
- [4] D. Zügner and S. Günnemann, “Adversarial attacks on graph neural networks via meta learning,” in *International Conference on Learning Representations*, 2019.
- [5] S. Geisler, T. Schmidt, H. Sirin, D. Zügner, A. Bojchevski, and S. Günnemann, “Robustness of graph neural networks at scale,” in *Advances in Neural Information Processing Systems*, vol. 34, 2021, pp. 7637–7649.
- [6] H. Wu, C. Wang, Y. Tyshetskiy, A. Docherty, K. Lu, and L. Zhu, “Adversarial examples on graph data: Deep insights into attack and defense,” in *International Joint Conference on Artificial Intelligence*, 2019, pp. 4816–4823.
- [7] A. Bojchevski, J. Klicpera, and S. Günnemann, “Efficient robustness certificates for discrete data: Sparsity-aware randomized smoothing for graphs, images and more,” in *International Conference on Machine Learning*, 2020, pp. 1003–1013.
- [8] A. Bojchevski and S. Günnemann, “Certifiable robustness to graph perturbations,” in *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [9] J. Schuchardt, Y. Scholten, A. Bojchevski, and S. Günnemann, “Localized randomized smoothing for GNNs,” in *International Conference on Machine Learning*, 2023.
- [10] J. Schuchardt, A. Bojchevski, J. Klicpera, and S. Günnemann, “Collective robustness certificates: Exploiting interdependence in graph neural networks,” in *International Conference on Learning Representations*, 2021.
- [11] Y. Li *et al.*, “AGNNcert: Deterministic certification for graph neural networks against adversarial perturbations,” in *International Conference on Learning Representations*, 2025.
- [12] D. Zügner and S. Günnemann, “Certifiable robustness and robust training for graph convolutional networks,” in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2019, pp. 246–256.
- [13] D. Zhu, Z. Zhang, P. Cui, and W. Zhu, “Robust graph convolutional networks against adversarial attacks,” in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2019, pp. 1399–1407.
- [14] X. Zhang and M. Zitnik, “GNNGuard: Defending graph neural networks against adversarial attacks,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9263–9275.
- [15] W. Jin, Y. Ma, X. Liu, X. Tang, S. Wang, and J. Tang, “Graph structure learning for robust graph neural networks,” in *Proceedings of the 26th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, 2020, pp. 66–74.
- [16] F. Mujkanovic, S. Geisler, S. Günnemann, and A. Bojchevski, “Are defenses for graph neural networks robust?” in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 8940–8952.
- [17] L. Gosch, S. Geisler, D. Sturm, and S. Günnemann, “Adversarial training for graph neural networks: Pitfalls, solutions, and new directions,” in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

- [18] P. W. Koh and P. Liang, "Understanding black-box predictions via influence functions," in *International Conference on Machine Learning*. PMLR, 2017, pp. 1885–1894.
- [19] S. Gould, R. Hartley, and D. Campbell, "Deep declarative networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 8, pp. 3988–4004, 2021.
- [20] S. Bai, J. Z. Kolter, and V. Koltun, "Deep equilibrium models," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [21] S. Bai, V. Koltun, and J. Z. Kolter, "Multiscale deep equilibrium models," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 5238–5250.
- [22] F. Gu, H. Chang, W. Zhu, S. Sojoudi, and L. El Ghaoui, "Implicit graph neural networks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 11 984–11 995.
- [23] T. Miyato, T. Kataoka, M. Koyama, and Y. Yoshida, "Spectral normalization for generative adversarial networks," in *International Conference on Learning Representations*, 2018.
- [24] S. W. Fung, H. Heaton, Q. Li, D. McKenzie, S. Osher, and W. Yin, "JFB: Jacobian-free backpropagation for implicit networks," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 6, 2022, pp. 6648–6656.
- [25] S. Bai, V. Koltun, and J. Z. Kolter, "Stabilizing equilibrium models by Jacobian regularization," in *International Conference on Machine Learning*, 2021, pp. 554–565.
- [26] L. N. Trefethen and M. Embree, *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*. Princeton University Press, 2005.
- [27] J. Lorraine, P. Vicol, and D. Duvenaud, "Optimizing millions of hyperparameters by implicit differentiation," in *International Conference on Artificial Intelligence and Statistics*, 2020, pp. 1540–1552.
- [28] Y. Sun, S. Wang, X. Tang, T.-Y. Hsieh, and V. Honavar, "Adversarial attacks on graph neural networks via node injections: A hierarchical reinforcement learning approach," in *Proceedings of The Web Conference (WWW)*, 2020.
- [29] L. El Ghaoui, F. Gu, B. Travacca, A. Askari, and A. Tsai, "Implicit deep learning," *SIAM Journal on Mathematics of Data Science*, vol. 3, no. 3, pp. 930–958, 2021.
- [30] M. Revay, R. Wang, and I. R. Manchester, "Lipschitz bounded equilibrium networks," in *arXiv preprint arXiv:2010.01732*, 2020.
- [31] J. Bolte and E. Pauwels, "Conservative set valued fields, automatic differentiation, stochastic gradient descent and deep learning," *Mathematical Programming*, vol. 188, pp. 19–51, 2021.
- [32] G. W. Stewart and J.-g. Sun, *Matrix Perturbation Theory*. Academic Press, 1990.
- [33] J. R. Magnus and H. Neudecker, *Matrix Differential Calculus with Applications in Statistics and Econometrics*, 3rd ed. Wiley, 2019.
- [34] N. Halko, P.-G. Martinsson, and J. A. Tropp, "Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions," *SIAM Review*, vol. 53, no. 2, pp. 217–288, 2011.
- [35] A. Paszke, S. Gross, F. Massa, A. Lerner, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [36] P. Sen, G. Namata, M. Bilgic, L. Getoor, B. Galligher, and T. Eliassi-Rad, "Collective classification in network data," *AI Magazine*, vol. 29, no. 3, pp. 93–106, 2008.
- [37] O. Shchur, M. Mumme, A. Bojchevski, and S. Günnemann, "Pitfalls of graph neural network evaluation," in *Relational Representation Learning Workshop, NeurIPS*, 2018.
- [38] P. Mernyei and M. Caron, "Wiki-CS: A Wikipedia-based benchmark for graph neural networks," in *arXiv preprint arXiv:2007.02901*, 2020.
- [39] Y. Dou, Z. Liu, L. Sun, Y. Deng, H. Peng, and P. S. Yu, "Enhancing graph neural network-based fraud detectors against camouflaged fraudsters," in *Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM)*, 2020, pp. 315–324.
- [40] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *International Conference on Learning Representations*, 2018.
- [41] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *International Conference on Learning Representations*, 2017.
- [42] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Advances in Neural Information Processing Systems*, 2017.
- [43] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, "How powerful are graph neural networks?" in *International Conference on Learning Representations*, 2019.
- [44] J. Klicpera, A. Bojchevski, and S. Günnemann, "Predict then propagate: Graph neural networks meet personalized PageRank," in *International Conference on Learning Representations*, 2019.
- [45] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *International Conference on Learning Representations*, 2018.
- [46] S. Brody, U. Alon, and E. Yahav, "How attentive are graph attention networks?" in *International Conference on Learning Representations*, 2022.
- [47] K. Xu, H. Chen, S. Liu, P.-Y. Chen, T.-W. Weng, M. Hong, and X. Lin, "Topology attack and defense for graph neural networks: An optimization perspective," in *International Joint Conference on Artificial Intelligence*, 2019, pp. 3961–3967.
- [48] Y. Abbahaddou, S. Ennadir, J. F. Lutzeyer, M. Vazirgiannis, and H. Boström, "Bounding the expected robustness of graph neural networks subject to node feature attacks," in *International Conference on Learning Representations (ICLR)*, 2024.
- [49] E. Winston and J. Z. Kolter, "Monotone operator equilibrium networks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 10 718–10 728.
- [50] R. Ying, D. Bourgeois, J. You, M. Zitnik, and J. Leskovec, "GNNExplainer: Generating explanations for graph neural networks," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [51] J. Topping, F. Di Giovanni, B. P. Chamberlain, X. Dong, and M. M. Bronstein, "Understanding over-squashing and bottlenecks on graphs via curvature," in *International Conference on Learning Representations*, 2022.
- [52] F. Di Giovanni, L. Giusti, F. Barbero, G. Luise, P. Lio, and M. M. Bronstein, "Over-squashing in graph neural networks: A comprehensive survey," in *International Conference on Machine Learning*, 2023.
- [53] F. Gama, J. Bruna, and A. Ribeiro, "Stability properties of graph neural networks," *IEEE Transactions on Signal Processing*, vol. 68, pp. 5680–5695, 2020.
- [54] H. Kenlay, D. Thanou, and X. Dong, "Stability and generalisation of graph neural networks via spectral analysis," in *International Conference on Machine Learning*, 2021, pp. 5412–5422.
- [55] B. Donon, B. Donnot, I. Guyon, and A. Marot, "Graph neural solver for power systems," in *International Joint Conference on Neural Networks*, 2019, pp. 1–8.
- [56] A. Varbella, B. Gjorgiev, and G. Sansavini, "Graph neural networks for fast contingency analysis of power systems," *IEEE Transactions on Power Systems*, 2024.